EXPERIMENT 2: STUDY OF BIPOLAR JUNCTION TRANSISTOR APPLICATION

INTRODUCTION

Small Signal Amplifier

An amplifier is an electronic circuit which is used to make the magnitude of the signal applied to its input bigger. In "Electronics", small signal amplifiers are commonly used circuits as they have the ability to amplify a relatively small input signal into a much larger output signal.

In order to amplify all of the input signal with distortion free, DC base biasing is required. The purpose is to establish a Q-point about which variations in current and voltage can occur in response to an AC signal. The biasing of a transistor is purely a DC operation. In applications where very small signal voltages must be amplified, variations about the Q-point are relatively small.

The transistor amplifier is non-linear and an incorrect bias setting will produce large amounts of distortion to the output waveform. Incorrect positioning of the Q-point on the load line will produce either Saturation Clipping or Cut-off Clipping. Thus it is desirable to set the Q-point of the amplifier half way along the load line.

After the transistor has been biased with the Q-point near the middle of the load line, we can put a small ac-voltage on the base (vin) (note that large input signal may produce large amounts of amplitude distortion due to clipping). That produces a large ac voltage at the collector (vout). This increase is called amplification.

The amount by which the amplifier "amplifies" the input signal is called gain. Gain is a ratio of output divided by input. Therefore, it has no units but is given the symbol (A): Voltage Gain (Av), Current Gain (Ai) and Power Gain (Ap). The gain of the amplifier can also be expressed in Decibels or simply dB.

OBJECTIVE

1. To apply basic laws, theorem and methods of analysis to design an amplifier circuit.
2. To verify the design parameters experimentally.

EQUIPMENT AND COMPONENT

1. BJT (2N3904)
2. Resistor
3. Capacitor
4. Vcc (common collector voltage) = 12V
5. AC signal, 50mV(p), 10kHz
6. 2 Channel Oscilloscope

CIRCUIT DIAGRAM

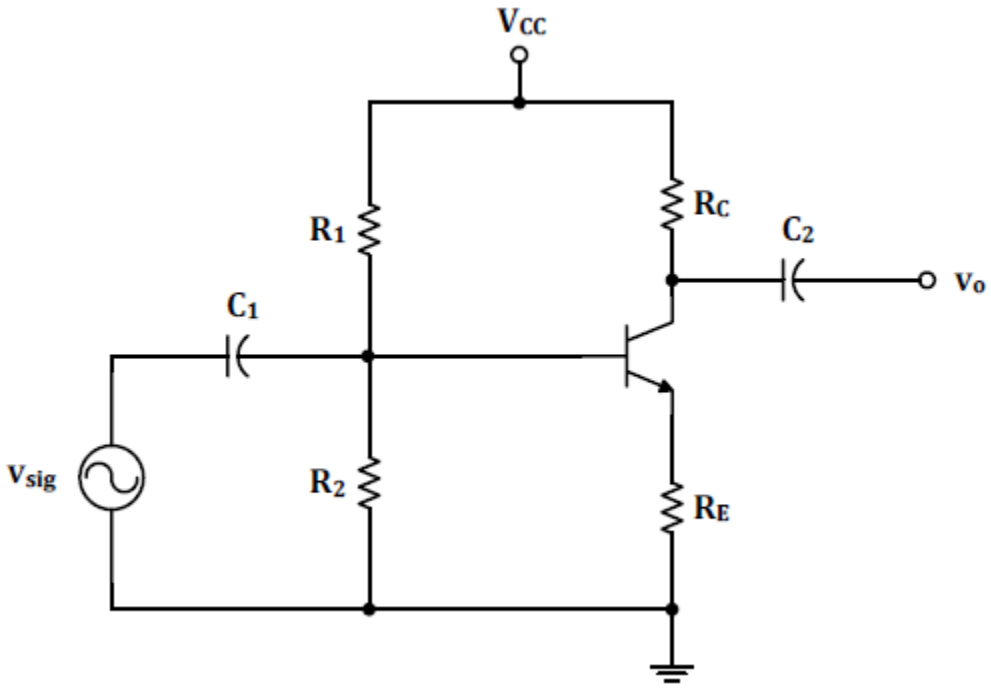


Figure 1

PART A: BJT as Small Signal Amplifier

1. Run Multisim 14.0 Software.
2. Click File > Save As > COM103_Experiment 2_YOUR ID_YOURNAME > Save.
3. Construct the circuit diagram as shown in figure 1 with the resistor value $R_1 = 27k\Omega$, $R_2 = 5k\Omega$, $R_c = 800\Omega$ and $R_E = 200\Omega$.
4. Click > place > component > basic > resistor > OK > place it on the screen.
5. Set the value of resistor as in step 3.
6. Click > place > component > basic > capacitor > OK > place it on the screen.
7. The capacitor values as listed in the Table 1.

Table 1: Capacitor values for the small-signal amplifier

Cycle	C_1 (µF)	C_2 (µF)	C_3 (µF)	C_4 (pF)
1	1	22	47	1000
2	2.2	47	33	2200
3	4.7	10	22	4700

8. Click > place > component > sources > signal voltage sources > ac voltage > OK > place it on the screen.
9. Set AC voltage > Voltage (pk) = 50mV > Frequency = 10 kHz > OK.
10. Click > place > component > transistors > BJT NPN > search > 2N3904 > OK > place it on the screen.
11. Connect the oscilloscope with the circuit input and output. Channel A to the input and Channel B to the output.
12. Run the simulation.
13. Measure the Vin signal and Vout signal from the output waveform and record the answer in Table 2.
14. Measure for the voltage gain using the equation (1).

$$A_v = \frac{V_o}{V_i}$$

Equation (1)

15. Repeat step 7 for the next cycle 2 and cycle 3. Record the value in Table 2.
16. Modify the circuit in figure 1 by adding C_3 and C_4 as shown in figure 2 below.

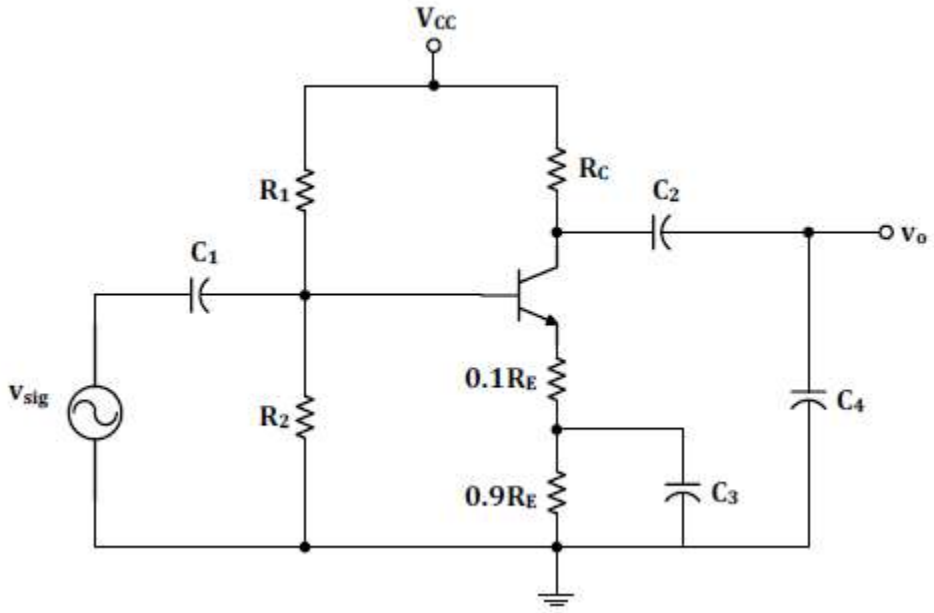


Figure 2

17. Repeat step 12 until step 14 and record the Vin signal and Vout signal from the output waveform in Table 3.
18. Discuss the effect of C_3 to the amplifier circuit.

PART B: BJT as a switch

EQUIPMENT AND COMPONENT

1. NPN transistor BC547BP
2. Resistor
3. DC source, 5V (Vcc and Vbb)
4. LED
5. Switch

CIRCUIT DIAGRAM

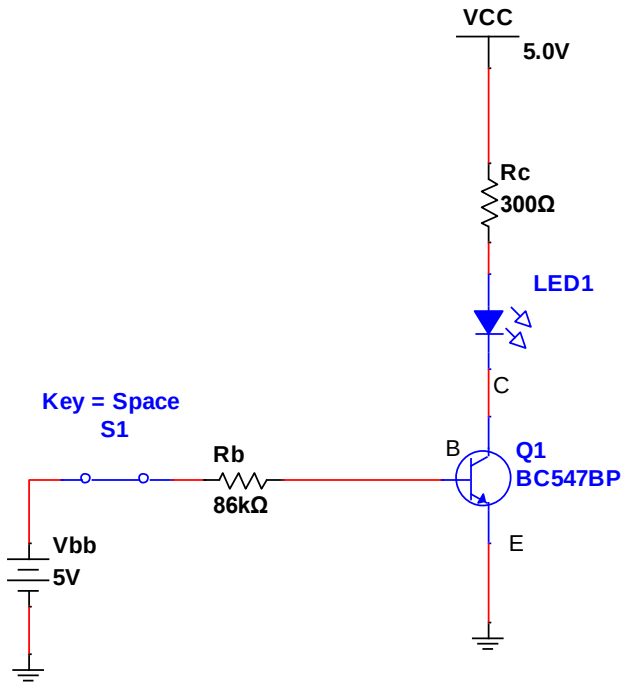


Figure 3

1. Run Multisim 14.0 Software.
2. Click File > Save As > COM103_Experiment 2_YOUR ID_YOURNAME > Save.

3. Click > place > component > transistors > BJT NPN > search > BC547BP > OK > place it on the screen.
4. Construct the circuit diagram as shown in figure 3 with the resistor value $R_b = 86$ kΩ and $R_c = 300$Ω.
5. Click > place > component > basic > resistor > OK > place it on the screen.
6. Set the value of resistor as in step 4.
7. Click > place > component > group > diodes > LED_red > OK > place it on the screen.
8. Click > place > component > basic > switch > SPST > OK > place it on the screen.
9. Click > place > component > group > sources > power sources > VCC > OK > place it on the screen.
10. Set Vcc = 5V.
11. Click > place > component > group > sources > power sources > DC power > OK > place it on the screen.
12. Set DC source = 5V.
13. Connect all component as in figure 3.
14. Run the simulation.
15. Measure the value of I_B, I_C, β, V_{BE}, V_{CE} (off), and V_{CE}(on) .
16. Record the answer in Table 4.

RESULT

PART A : BJT as amplifier

Table 2: Response of the circuit of figure 1

Cycle	$C_{1(uF)}$	$C_{2(uF)}$	$V_{in(mV)}$	$V_{out(mV)}$	A_V
1					
2					
3					

Table 3: Response of the circuit of figure 2

Cycle	$C_{1(uF)}$	$C_{2(uF)}$	$C_{3(uF)}$	$C_{4(pF)}$	$V_{in(mV)}$	$V_{out(mV)}$	A_V
1							
2							
3							

PART B: BJT as a Switch

Table 4: Response of the circuit of figure 3

Parameter	*measured value*	*calculated value*
I_B		
Ic		
β		
$V_{BE}(on)$		
$V_{BE}(off)$		
$V_{CE}\ (on)$		
$V_{CE}(off)$		

$$\beta = \frac{I_C}{I_B}$$

Discussion

1. What are the characteristics of the amplifier that you can observe in Part A?
2. Write simple statement about the effect of C_3 on the magnitude of voltage gain. Used your measured value to back up your statement.
3. What is the mode of operation for V_{BE}, V_{CE} in ON OFF condition in Part B?
4. Show all calculation for Table 2, 3 and 4.